



FAIR 3D HERITAGE

*Documentation, Publication and Preservation
of the Digital 3D Heritage on the Web*

WebHDP : A FAIR-Oriented Publication Workflow for 3D Heritage Data

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Abstract

Objective and context. Three-dimensional cultural heritage projects combine geometry, textures, images, tabular metadata, and documentation. Publication still often depends on manual preparation and ad hoc scripts, weakening reproducibility and the relationship between deposited objects and their documentation. The FAIR principles require data and metadata to be Findable, Accessible, Interoperable, and Reusable, not merely downloadable (Wilkinson et al., 2016; Jacobsen et al., 2020). Preservation guidance likewise stresses retaining relationships among 3D sources, derivatives, metadata, and publication targets (Amico and Felicetti, 2024). This contribution examines how WebHDP implements FAIR-enabling controls across these stages in a workflow usable by institutional teams without expertise in every underlying protocol.

Methods and data. WebHDP is an open-source, containerised web application with public source code and documentation (WebHDP Project, 2026). A representative 3D record comprises a GLB source with related assets and one row in a CSV or XLSX metadata spreadsheet file (MSF). Files, checksums, mappings, processing events, and record identities persist in a resumable **.hdpc** session database. Before processing, WebHDP reports filename-column confidence, blocks non-deterministic duplicate filenames, and identifies missing or unmapped files. Users select the authoritative source for each target field rather than allowing the system to arbitrate contradictory metadata. Zenodo, EDM, IIIF, and METS/MODS representations are previewed and validated separately. Filename-derived or default fallbacks satisfy only minimum structural requirements and remain visible warnings requiring curatorial acceptance.

Implementation and achieved results. The current alpha completes an end-to-end record workflow: it can render front, side, back, and top views, convert a 3D source to USDZ, generate Europeana Data Model (EDM) XML, an IIIF Presentation manifest, and a METS document containing MODS descriptive metadata (Isaac, 2013; Appleby et al., 2020; METS Editorial Board, 2010), and manage Zenodo drafts, publication, and versioning. Source registration, derivatives, validation reports, and repository events remain linked in the same session. This provides concrete FAIR alignment: published records can receive version and concept DOIs with rich metadata; DOI/HTTP retrieval and explicit access conditions support accessibility; RDF-based EDM, IIIF JSON-LD, METS/MODS XML, and controlled vocabulary URIs support interoperability; and licences, checksums, source-to-derivative lineage, metadata-source mappings, and version history support reuse. Cross-session hash comparison identifies duplicate, new-version, and conflicting-lineage candidates.

User, moderator, and administrator roles, review queues, and recorded override reasons provide an auditable governance layer. These are FAIR-enabling controls, not a claim of automatic FAIR compliance.

Relevance and near-term work. WebHDP directly addresses the conference themes of documentation practice and web-based publication and preservation infrastructure by making repository submission reviewable rather than a final upload step. Proposed, not yet implemented, Small Language Model assistance would suggest validation-grounded metadata repairs for human approval; Retrieval-Augmented Generation would support access-controlled queries with links to record and provenance evidence. Future evaluation will compare correction accuracy, unsupported suggestions, and task completion time against the existing manual workflow.

References

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